Heat Transfer

MEP 212

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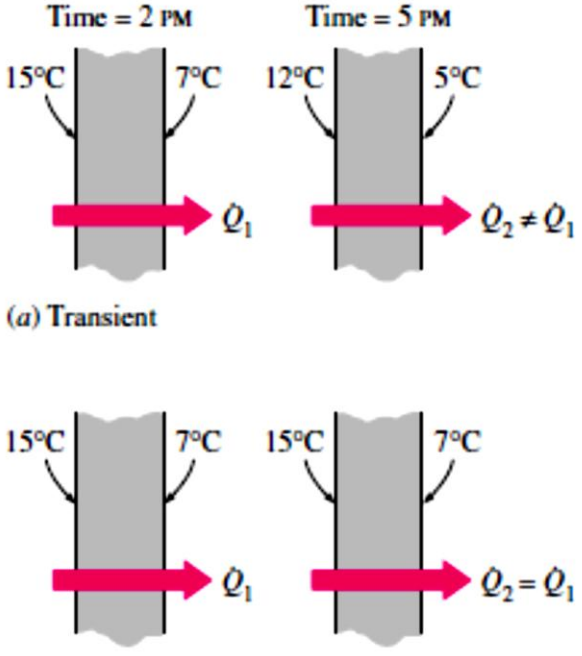
Steady state 1-Dimensional Conduction Heat transfer

Steady vs. Transient

- The term ***steady*** implies *no change* with time at any point within the medium, while ***transient*** implies *variation with time* or *time dependence.*
- Therefore, the temperature or heat flux remains unchanged with time during steady heat transfer through a medium at any location, although both quantities may vary from one location to another



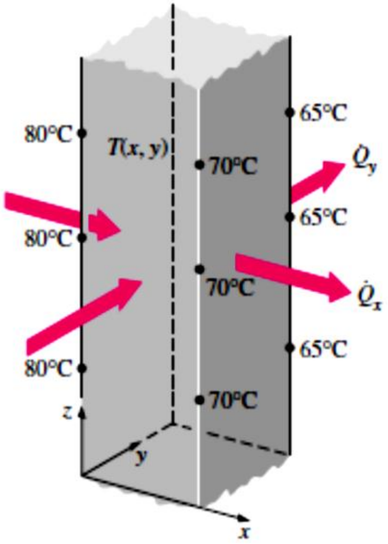
(a) Transient

(b) Steady-state

Multi-dimensional conduction

Heat transfer problems are also classified as being *one-dimensional, two dimensional,* or *three-dimensional,* depending on the relative magnitudes of heat transfer rates in different directions and the level of accuracy desired. In the most general case, heat transfer through a medium is **three-dimensional.**

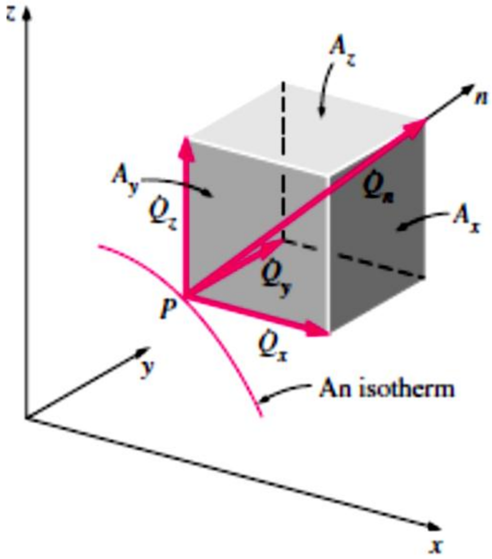
That is, the temperature varies along all three primary directions within the medium during the heat transfer process.



Multi-dimensional conduction

$$\vec{\dot{Q}}_n = \dot{Q}_x \vec{i} + \dot{Q}_y \vec{j} + \dot{Q}_z \vec{k}$$

$$\dot{Q}_x = -kA_x \frac{\partial T}{\partial x}, \qquad \dot{Q}_y = -kA_y \frac{\partial T}{\partial y}, \qquad \text{and} \qquad \dot{Q}_z = -kA_z \frac{\partial T}{\partial z}$$

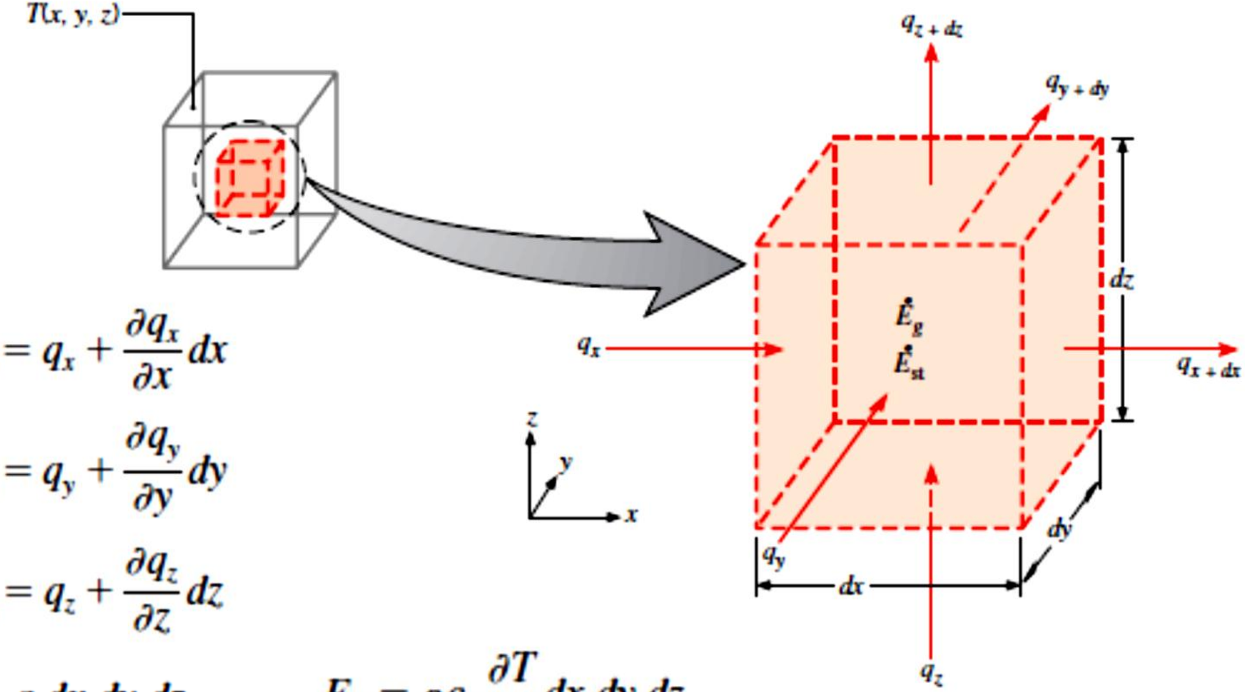


Heat generation

- A medium through which heat is conducted may involve the conversion of electrical, nuclear, or chemical energy into heat (or thermal) energy. In heat conduction analysis, such conversion processes are characterized as **heat generation.**
- For example, the temperature of a resistance wire rises rapidly when electric current passes through it as a result of the electrical energy being converted to heat at a rate of I^2R, where I is the current and R is the electrical resistance of the wire

$$\dot{G} = \int_V \dot{g}\, dV$$

General heat conduction equation



$$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx$$

$$q_{y+dy} = q_y + \frac{\partial q_y}{\partial y} dy$$

$$q_{z+dz} = q_z + \frac{\partial q_z}{\partial z} dz$$

$$E_g = q\, dx\, dy\, dz \qquad E_{st} = \rho c_p \frac{\partial T}{\partial t} dx\, dy\, dz$$

General heat conduction equation

$$E_{in} + E_g - E_{out} = E_{st}$$

$$q_x + q_y + q_z + q\,dx\,dy\,dz - q_{x+dx} - q_{y+dy} - q_{z+dz} = \rho c_p \frac{\partial T}{\partial t} dx\,dy\,dz$$

$$-\frac{\partial q_x}{\partial x}dx - \frac{\partial q_y}{\partial y}dy - \frac{\partial q_z}{\partial z}dz + q\,dx\,dy\,dz = \rho c_p \frac{\partial T}{\partial t} dx\,dy\,dz$$

$$q_x = -k\,dy\,dz\frac{\partial T}{\partial x}$$

$$q_y = -k\,dx\,dz\frac{\partial T}{\partial y}$$

$$q_z = -k\,dx\,dy\frac{\partial T}{\partial z}$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + q = \rho c_p \frac{\partial T}{\partial t}$$

General heat conduction equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

Constant thermal conductivity

where $\alpha = k/\rho c_p$ is the *thermal diffusivity.*

The thermal diffusivity, which represents how fast heat diffuses through a material and is defined as

$$\alpha = \frac{\text{Heat conducted}}{\text{Heat stored}} = \frac{k}{\rho C_p} \qquad (\mathrm{m^2/s})$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + q = 0$$

Steady State conditions

$$\frac{d}{dx}\left(k\frac{dT}{dx}\right) = 0$$

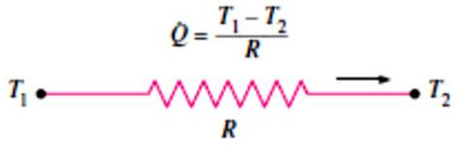
One dimensional, Steady State conduction with no generation

Steady heat conduction- plane wall

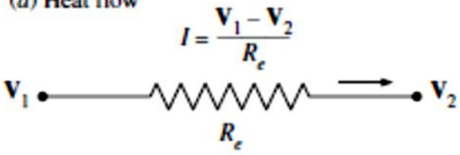
$$\dot{Q}_{\text{cond, wall}} = kA\frac{T_1 - T_2}{L}$$

$$\dot{Q}_{\text{cond, wall}} = \frac{T_1 - T_2}{R_{\text{wall}}} \qquad \text{(W)}$$

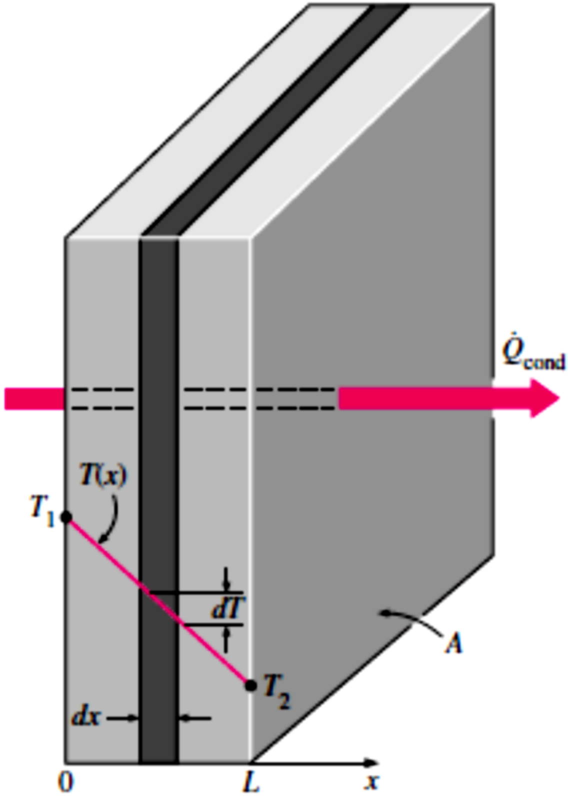
$$R_{\text{wall}} = \frac{L}{kA} \qquad (^{\circ}\text{C/W})$$



(a) Heat flow



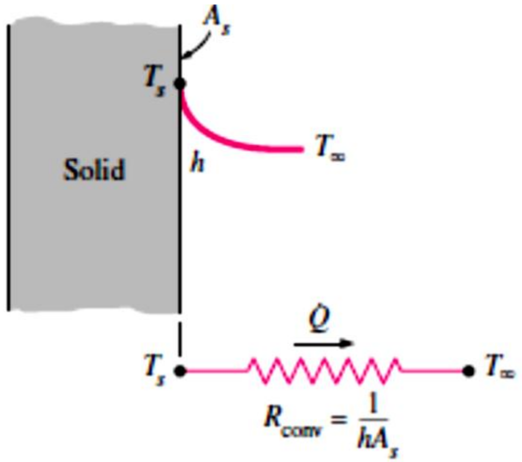
(b) Electric current flow



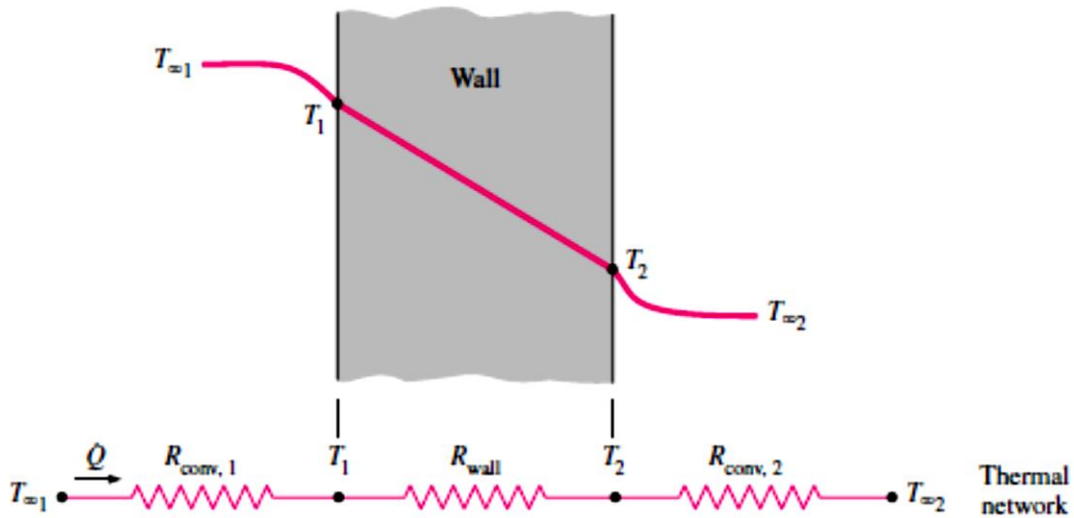
Steady heat conduction- plane wall

$$\dot{Q}_{conv} = \frac{T_s - T_\infty}{R_{conv}} \qquad \text{(W)}$$

$$R_{conv} = \frac{1}{hA_s} \qquad \text{(°C/W)}$$

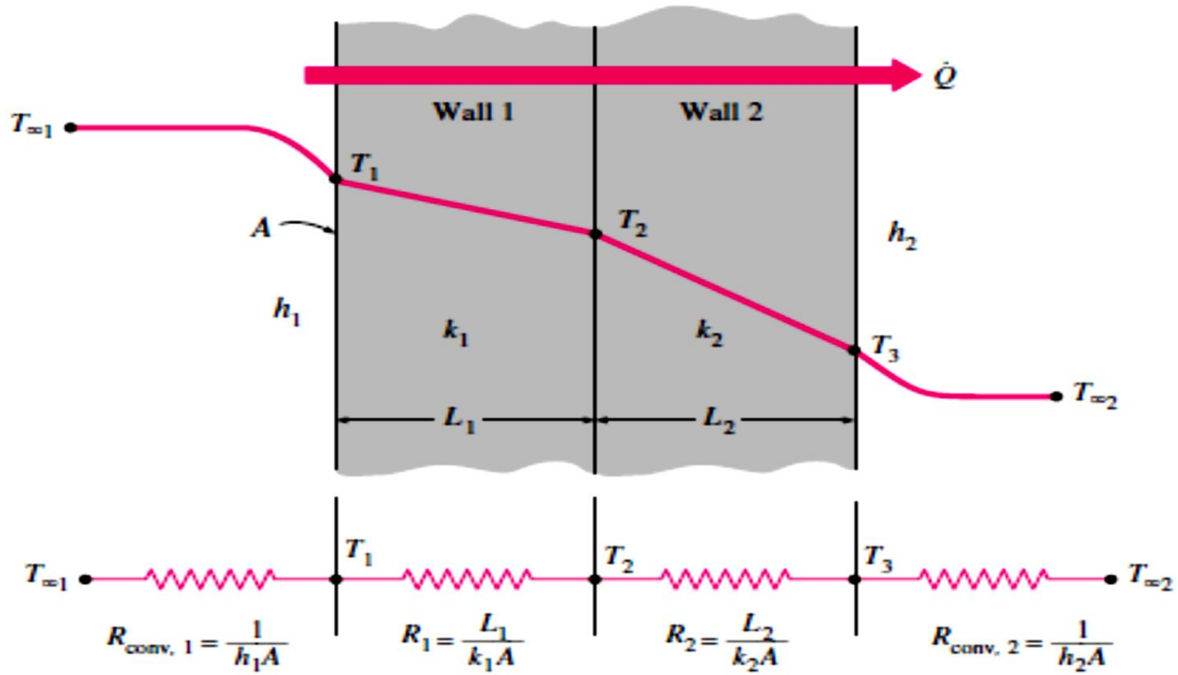


Steady heat conduction- plane wall



$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{conv,\,1} + R_{wall} + R_{conv,\,2}}$$

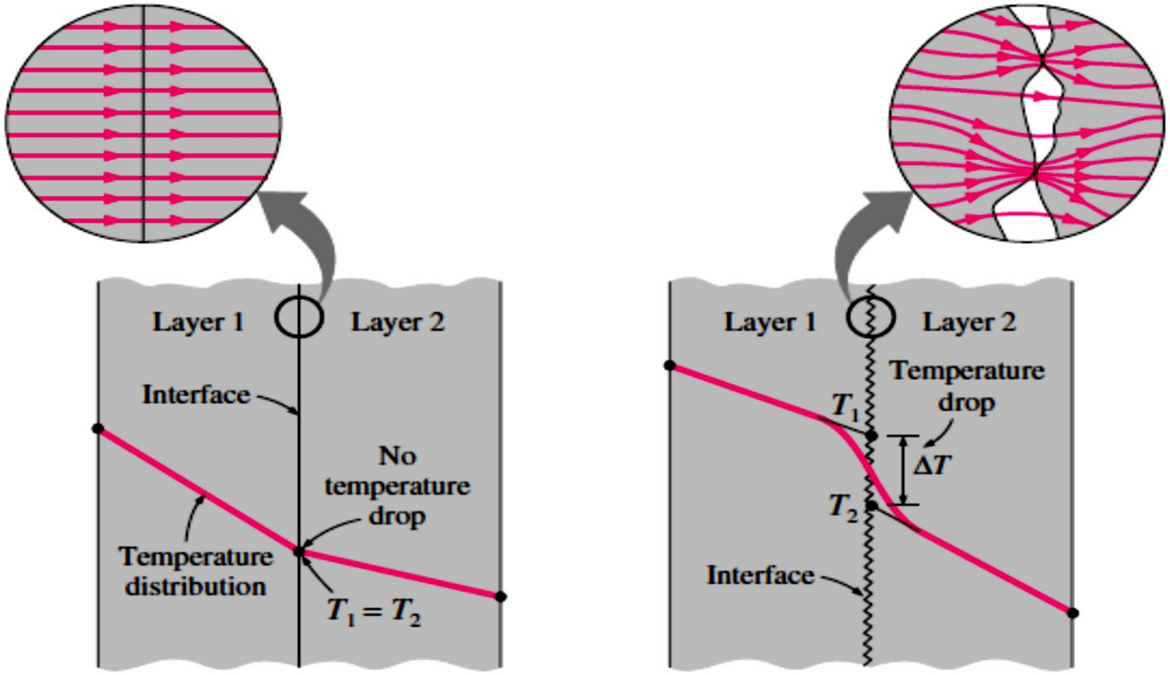
Steady heat conduction- composite wall



Thermal contact resistance

In the analysis of heat conduction through multilayer solids, we assumed “perfect contact” at the interface of two layers, and thus no temperature drop at the interface. This would be the case when the surfaces are perfectly smooth and they produce a perfect contact at each point. In reality, however, even flat surfaces that appear smooth to the eye turn out to be rather rough when examined under a microscope with numerous peaks and valleys.

Thermal contact resistance



(*a*) Ideal (perfect) thermal contact

(*b*) Actual (imperfect) thermal contact

Thermal contact resistance

When two such surfaces are pressed against each other, the peaks will form good material contact but the valleys will form voids filled with air. As a result, an interface will contain numerous *air gaps* of varying sizes that act as *insulation* because of the low thermal conductivity of air. Thus, an interface offers some resistance to heat transfer, and this resistance per unit interface area is called the **thermal contact resistance,** *Rc*.

Factors affecting the contact resistance

- Surface roughness
- Material properties
- Temperature and pressure at the interface
- Type of fluid trapped in the interface

Decreasing the contact resistance

- Decreasing the surface roughness
- Increasing interface pressure
- applying a thermally conducting liquid called a *thermal grease* such as silicon oil on the surfaces before they are pressed against each other.
- insert a *soft metallic foil* such as tin, silver, copper, nickel, or aluminum between the two surfaces.

Parallel resistance network

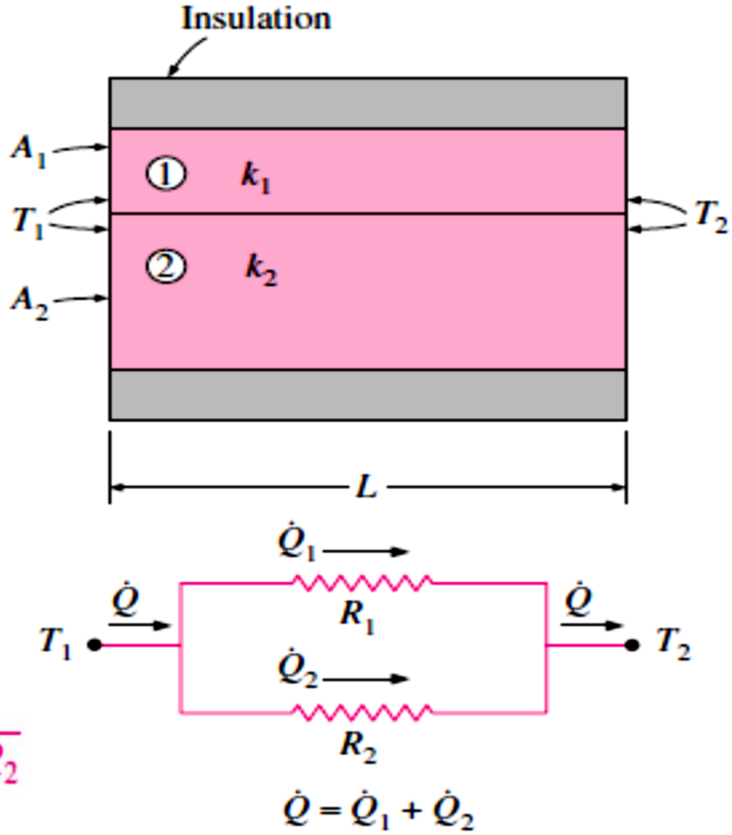
$$\dot{Q} = \dot{Q}_1 + \dot{Q}_2$$

$$= \frac{T_1 - T_2}{R_1} + \frac{T_1 - T_2}{R_2}$$

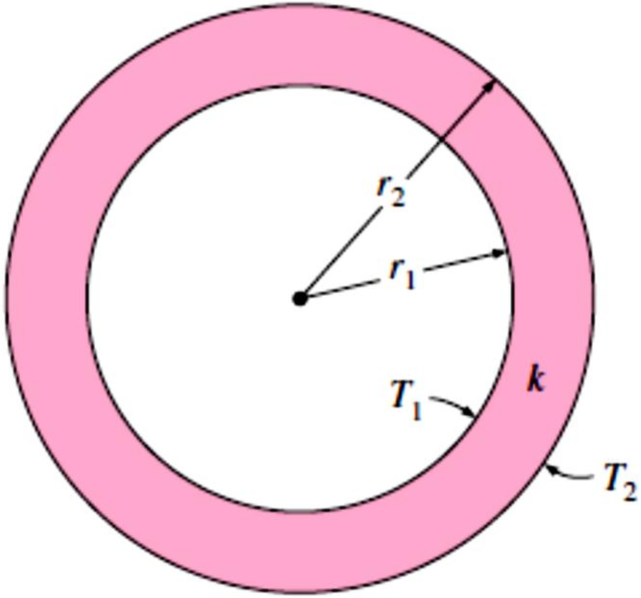
$$= (T_1 - T_2)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

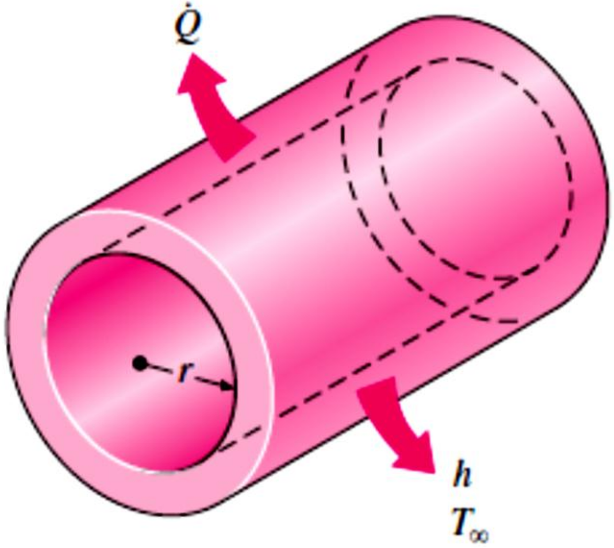
$$\dot{Q} = \frac{T_1 - T_2}{R_{\text{total}}}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} \longrightarrow R_{\text{total}} = \frac{R_1 R_2}{R_1 + R_2}$$



Heat conduction in cylinders and spheres



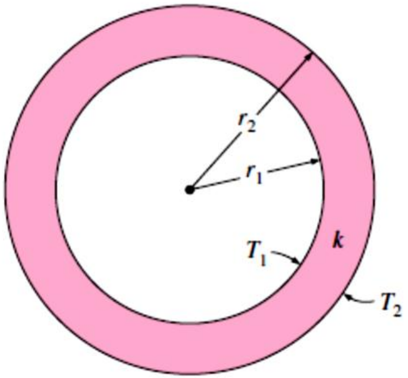


Cylindrical walls

$$\dot{Q}_{\text{cond, cyl}} = -kA\frac{dT}{dr} \qquad \text{(W)}$$

$$A = 2\pi rL$$

$$\int_{r=r_1}^{r_2} \frac{\dot{Q}_{\text{cond, cyl}}}{A}\,dr = -\int_{T=T_1}^{T_2} k\,dT$$



Substituting $A = 2\pi rL$ and performing the integrations give

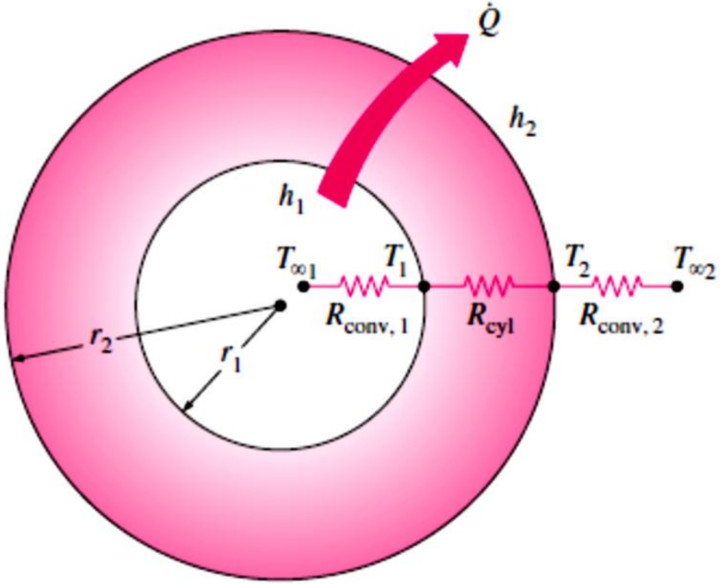
$$\dot{Q}_{\text{cond, cyl}} = 2\pi Lk\frac{T_1 - T_2}{\ln(r_2/r_1)} \qquad \text{(W)}$$

$$\dot{Q}_{\text{cond, cyl}} = \frac{T_1 - T_2}{R_{\text{cyl}}} \qquad R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi Lk}$$

Cylindrical walls

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{total}}$$

$$R_{total} = R_{conv,1} + R_{cyl} + R_{conv,2} = \frac{1}{(2\pi r_1 L)h_1} + \frac{\ln(r_2/r_1)}{2\pi L k} + \frac{1}{(2\pi r_2 L)h_2}$$



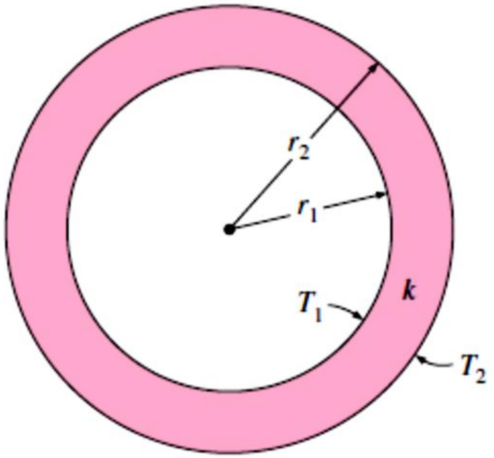
$$R_{total} = R_{conv,1} + R_{cyl} + R_{conv,2}$$

Spherical walls

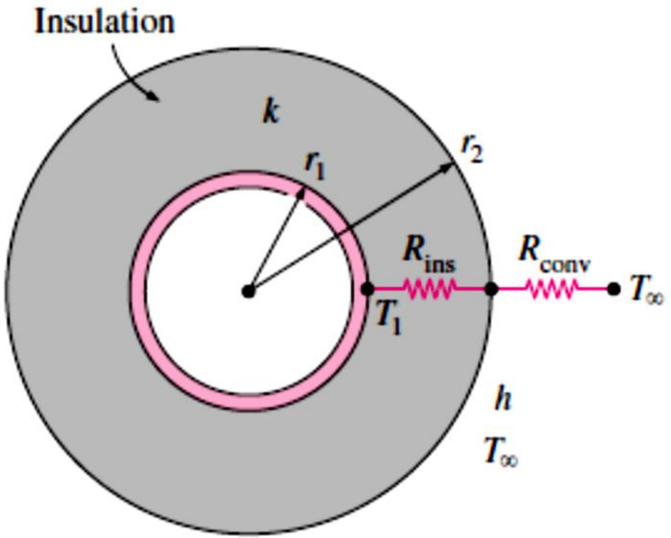
$$A = 4\pi r^2$$

$$\dot{Q}_{\text{cond, sph}} = \frac{T_1 - T_2}{R_{\text{sph}}}$$

$$R_{\text{sph}} = \frac{r_2 - r_1}{4\pi r_1 r_2 k}$$



Critical radius of insulation

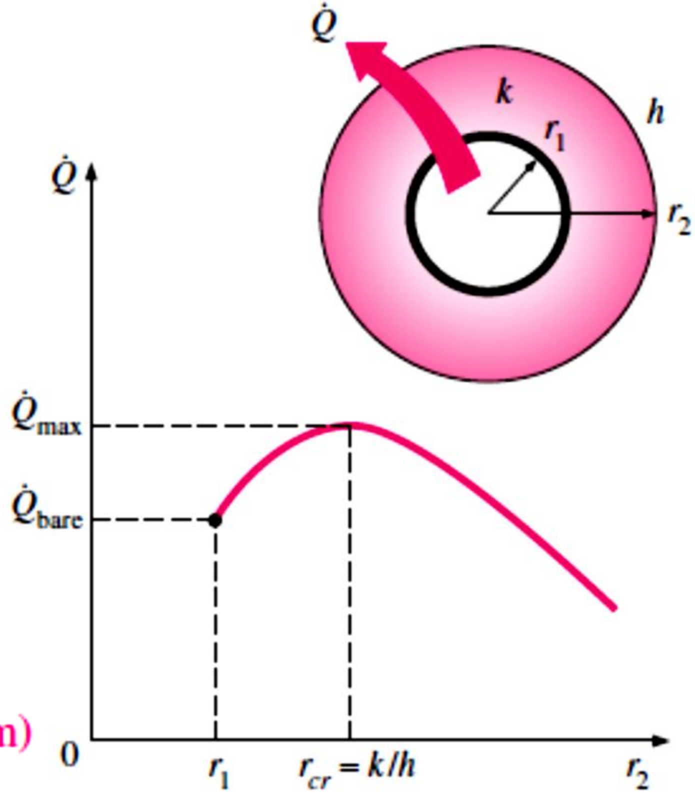


$$\dot{Q} = \frac{T_1 - T_\infty}{R_{ins} + R_{conv}} = \frac{T_1 - T_\infty}{\dfrac{\ln(r_2/r_1)}{2\pi Lk} + \dfrac{1}{h(2\pi r_2 L)}}$$

Critical radius of insulation

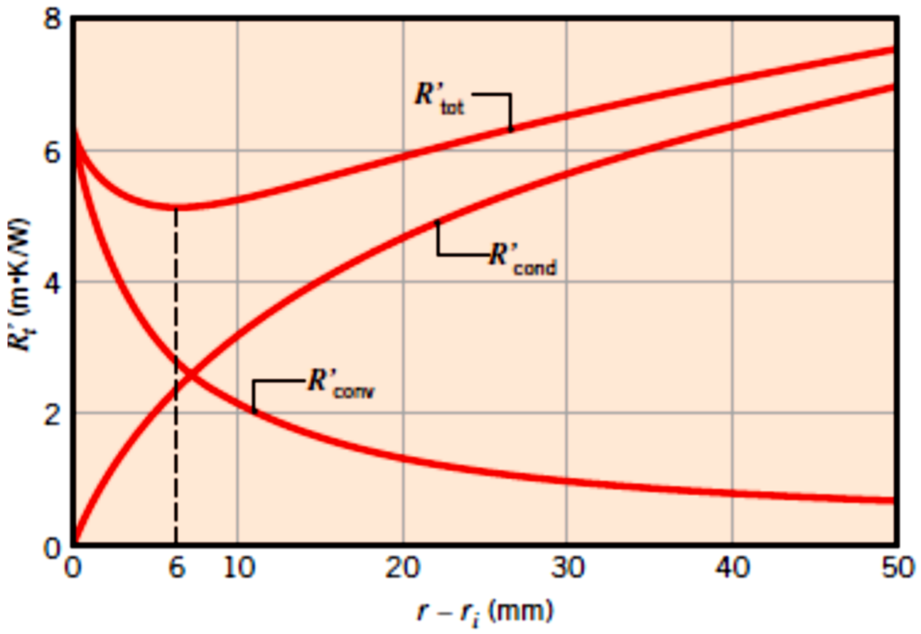
The variation of *Q'* with the outer radius of the insulation r_2 is plotted in the figure. The value of r_2 at which *Q'* reaches a maximum is determined from the requirement that $dQ'/d\,r_2 = 0$ (zero slope). Performing the differentiation and solving for r_2 yields the **critical radius of insulation** for a cylindrical bo

$$r_{\text{cr, cylinder}} = \frac{k}{h} \qquad \text{(m)}$$



Critical radius of insulation

$$r_{\text{cr, sphere}} = \frac{2k}{h}$$



Temperature distribution

Plane wall

$$\frac{d}{dx}\left(k\frac{dT}{dx}\right) = 0$$

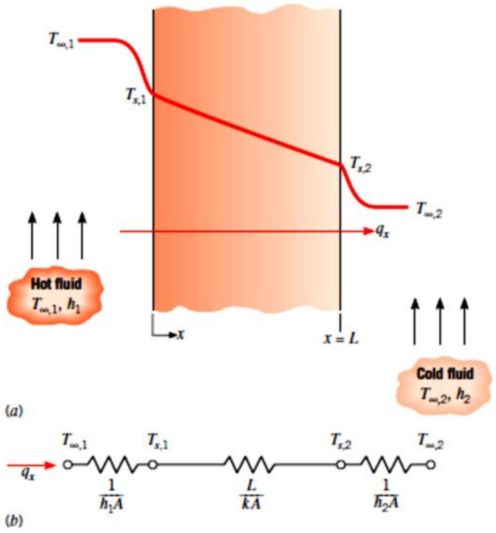
$$T(x) = C_1 x + C_2$$

$$T(0) = T_{s,1} \quad \text{and} \quad T(L) = T_{s,2}$$

$$T_{s,1} = C_2 \qquad T_{s,2} = C_1 L + C_2 = C_1 L + T_{s,1}$$

$$\frac{T_{s,2} - T_{s,1}}{L} = C_1$$

$$T(x) = (T_{s,2} - T_{s,1})\frac{x}{L} + T_{s,1}$$



Temperature distribution

Cylindrical wall

$$\frac{1}{r}\frac{d}{dr}\left(kr\frac{dT}{dr}\right)=0$$

$$T(r) = C_1 \ln r + C_2$$

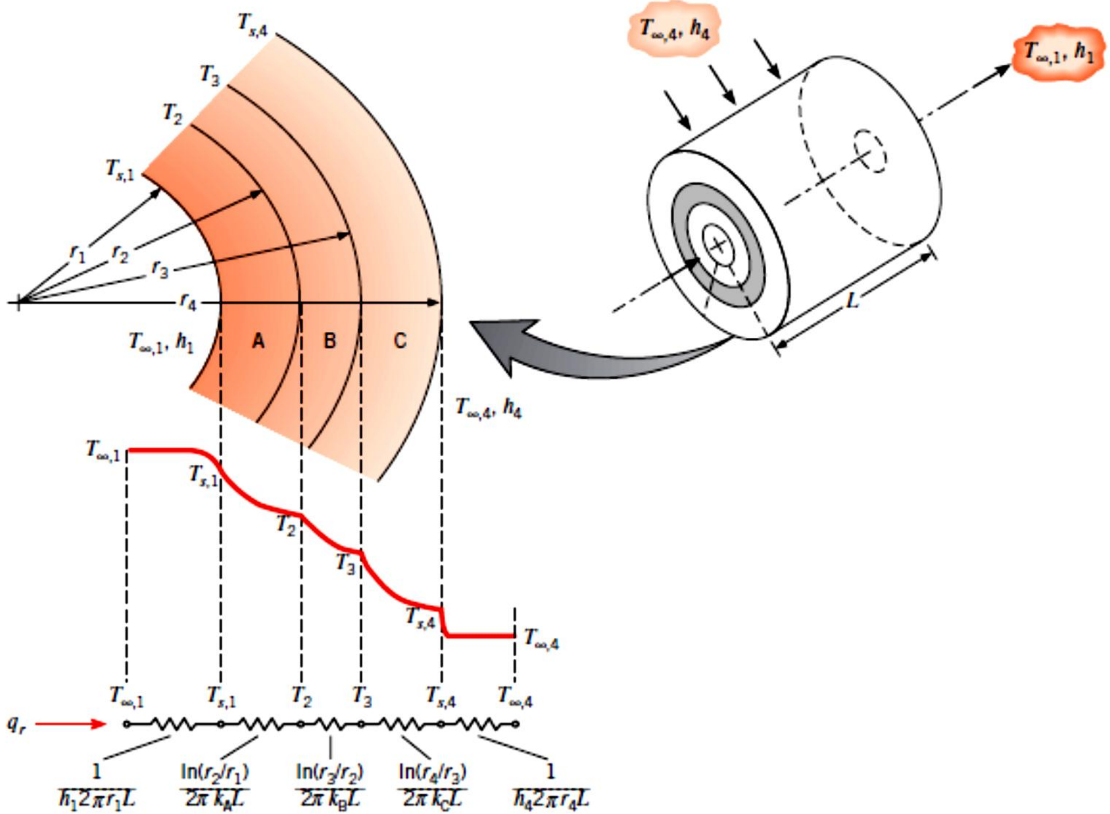
$$T(r_1) = T_{s,1} \quad \text{and} \quad T(r_2) = T_{s,2}$$

$$T_{s,1} = C_1 \ln r_1 + C_2 \quad \text{and} \quad T_{s,2} = C_1 \ln r_2 + C_2$$

$$T(r) = \frac{T_{s,1} - T_{s,2}}{\ln(r_1/r_2)} \ln\left(\frac{r}{r_2}\right) + T_{s,2}$$

Temperature distribution

Cylindrical wall



Conclusion

TABLE 3.3 One-dimensional, steady-state solutions to the heat equation with no generation

	Plane Wall	Cylindrical Wall[a]	Spherical Wall[a]
Heat equation	$\frac{d^2T}{dx^2} = 0$	$\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) = 0$	$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dT}{dr}\right) = 0$
Temperature distribution	$T_{s,1} - \Delta T\frac{x}{L}$	$T_{s,2} + \Delta T\frac{\ln(r/r_2)}{\ln(r_1/r_2)}$	$T_{s,1} - \Delta T\left[\frac{1-(r_1/r)}{1-(r_1/r_2)}\right]$
Heat flux (q'')	$k\frac{\Delta T}{L}$	$\frac{k\,\Delta T}{r\ln(r_2/r_1)}$	$\frac{k\,\Delta T}{r^2[(1/r_1)-(1/r_2)]}$
Heat rate (q)	$kA\frac{\Delta T}{L}$	$\frac{2\pi Lk\,\Delta T}{\ln(r_2/r_1)}$	$\frac{4\pi k\,\Delta T}{(1/r_1)-(1/r_2)}$
Thermal resistance ($R_{t,cond}$)	$\frac{L}{kA}$	$\frac{\ln(r_2/r_1)}{2\pi Lk}$	$\frac{(1/r_1)-(1/r_2)}{4\pi k}$

[a]The critical radius of insulation is $r_{cr} = k/h$ for the cylinder and $r_{cr} = 2k/h$ for the sphere.

Thank you